Algorithms need to be tested using the same kind of data they will encounter in actual operation. This creates the need to generate digital noise with a *Gaussian* pdf. There are two methods for generating such signals using a random number generator. Figure 2-10 illustrates the first method. Figure (b) shows a signal obtained by adding two random numbers to form each sample, i.e., $X = RND+RND$. Since each of the random numbers can run from zero to one, the sum can run from zero to two. The mean is now *one*, and the standard deviation is $1/\sqrt{6}$ (remember, when independent random signals are added, the variances also add). As shown, the pdf has changed from a *uniform* distribution to a *triangular* distribution. That is, the signal spends more of its time around a value of *one*, with less time spent near *zero* or *two*.

Figure (c) takes this idea a step further by adding twelve random numbers to produce each sample. The mean is now *six*, and the standard deviation is *one*. What is most important, the pdf has virtually become a *Gaussian*. This procedure can be used to create a normally distributed noise signal with an arbitrary mean and standard deviation. For each sample in the signal: (1) add twelve random numbers, (2) subtract six to make the mean equal to zero, (3) multiply by the standard deviation desired, and (4) add the desired mean.

The mathematical basis for this algorithm is contained in the **Central Limit Theorem**, one of the most important concepts in probability. In its simplest form, the Central Limit Theorem states that a *sum* of random numbers becomes normally distributed as more and more of the random numbers are added together. The Central Limit Theorem *does not* require the individual random numbers be from any particular distribution, or even that the random numbers be from the *same* distribution. The Central Limit Theorem provides the reason why normally distributed signals are seen so widely in nature. Whenever many different random forces are interacting, the resulting pdf becomes a Gaussian.

In the second method for generating normally distributed random numbers, the random number generator is invoked *twice*, to obtain R_1 and R_2. A normally distributed random number, X, can then be found:

EQUATION 2-9
Generation of normally distributed random numbers. R_1 and R_2 are random numbers with a uniform distribution between zero and one. This results in X being normally distributed with a mean of zero, and a standard deviation of one. The log is base *e*, and the cosine is in radians.

$$X = (-2 \log R_1)^{1/2} \cos(2\pi R_2)$$

Just as before, this approach can generate normally distributed random signals with an arbitrary mean and standard deviation. Take each number generated by this equation, multiply it by the desired standard deviation, and add the desired mean.